

5.2.1 Scope of AI in Education in Pakistan

پاکستان کے تعلیمی نظام میں مصنوعی ذہانت آہستہ آہستہ اہم کردار ادا کر رہی ہے، جیسے لرننگ مینجمنٹ سسٹم (LMS)،
خو...

CS XI

Chapter 5: Application and Impacts of Computing

Lecture 07 of 13

5.2.1 Scope of AI in Education in Pakistan

LEARN IT

Artificial Intelligence is gradually becoming an important part of Pakistan's education system, improving **teaching, learning** and **administration** (managing the school: admissions, records, timetables).

Practical uses:

- **Learning Management Systems (LMS)** — online platforms (e.g. Google Classroom, Moodle) where teachers post lessons and assignments; AI features can recommend lessons to each student.
- **Automated grading and digital testing** — MCQ tests marked instantly by computer.
- **AI-based career counselling** — software that studies a student's marks and interests and suggests suitable fields.

Benefits:

- Helps students learn **at their own speed**.
- **Identifies students who need extra help**.
- Supports students with disabilities (**Assistive AI**), e.g. text-to-speech for a visually impaired student.

Major challenges in Pakistan:

- Limited **internet access in rural areas**.
- Lack of **computers and digital devices** for all students.
- Need for proper **teacher training** to use AI tools.
- **Data privacy and security** issues — students' personal data must be protected.

Practical uses	Benefits	Challenges
LMS	Learn at own speed	Limited rural internet
Automated grading & digital testing	Identifies students needing help	Lack of computers/devices
AI career counselling	Supports students with disabilities	Teacher training needed
—	—	Data privacy & security

WORKED EXAMPLE

Scenario — Fatima in Larkana and an AI tutor app:

Step 1. Fatima takes a 10-question online maths quiz on the college LMS. It is graded instantly (automated grading).

Step 2. The AI notices she got all 3 trigonometry questions wrong.

Step 3. It recommends two short videos and five easy practice questions on trigonometry — she learns at her own speed.

Step 4. Her teacher's dashboard flags "Fatima – needs help in trigonometry" (identifies students who need help).

Challenge: At home Fatima's village has weak 3G signals, so the videos buffer — showing the rural internet challenge.

COMMON MISTAKE

Mistake: Writing that AI will replace teachers. **Fix:** The source says AI *improves* teaching and needs *teacher training* — teachers use AI as a tool; they are not replaced.

اردو میں سمجھیں

پاکستان کے تعلیمی نظام میں مصنوعی ذہانت آہستہ آہستہ اہم کردار ادا کر رہی ہے، جیسے لرننگ مینجمنٹ سسٹم (LMS)، خودکار جانچ اور کیریئر کونسلنگ۔ اس سے طلبہ اپنی رفتار سے سیکھ سکتے ہیں اور کمزور طلبہ کی نشاندہی ہوتی ہے۔ لیکن دیہی علاقوں میں انٹرنیٹ کی کمی، کمپیوٹرز کی قلت، اساتذہ کی تربیت اور ڈیٹا کی حفاظت بڑے چیلنج ہیں۔

CHECK YOURSELF

1. Give three practical uses of AI in education.
2. State any two challenges of AI in education in Pakistan.

Answers: 1. LMS, automated grading and digital testing, AI-based career counselling. 2. Any two: limited rural internet, lack of devices, need for teacher training, data privacy and security.

Next lecture: 5.2.2 Career Connection in IoT and AI